| Question | Answer | Marks |
|---|---|---|
| 1 | **Defining the problem** | |
| | *s* is the independent variable and *v* is the dependent variable **or** vary *s* and measure *v* | **1** |
| | keep *D* constant | **1** |
| | **Methods of data collection** | |
| | labelled diagram of workable experiment including:<br>• light gate positioned at P<br>• light gate connected to timer / data logger<br>• labels for light gate **and** P **and** data logger / timer **and** at least one other label from block, magnet(s), trolley, *s* and *D* | **1** |
| | measure *D* with a rule(r) **and** measure *L* with a rule(r) or calipers | **1** |
| | description to determine *v* at P, e.g. (measure length of) card to interrupt beam | **1** |
| | method to measure *s*, e.g. use calipers | **1** |

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|---|---|---|
| 1 | **Method of Analysis** | |
| | plot a graph of $v^2$ against $\frac{1}{s^4}$ or equivalent (e.g. $\frac{1}{s^4}$ against $v^2$)<br><br>Do not accept logarithms. | **1** |
| | $K = \frac{m \times \text{gradient}}{2DA^2B^2L^2}$<br><br>(or $K = \frac{m}{2DA^2B^2L^2 \times \text{gradient}}$ for $\frac{1}{s^4}$ against $v^2$) | **1** |
| | $Q = -\frac{m \times y\text{-intercept}}{2D}$<br><br>(or $Q = KA^2B^2L^2 \times y\text{-intercept}$ or $Q = \frac{m \times y\text{-intercept}}{2D \times \text{gradient}}$ for $\frac{1}{s^4}$ against $v^2$) | **1** |

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|---|---|---|
| 1 | **Additional detail including safety considerations** | **6** |
| | D1 method to stop the trolley (after passing point P), e.g. labelled block / buffer / cushion drawn after P<br>**or**<br>place a block / buffer / cushion after P to stop the trolley | |
| | D2 keep *L*, *A*, *m* and *B* constant | |
| | D3 use micrometer / calipers to measure diameter (*d*) of the magnet **and** $A = \frac{\pi d^2}{4}$ | |
| | D4 method to secure block to bench, e.g. clamp block to bench or (heavy) mass on top of block<br>**or**<br>method to secure magnets, e.g. use glue to stick magnets to trolley / block | |
| | D5 method to increase the accuracy of measuring *s* or *D*, e.g. use a marker to left of the trolley | |
| | D6 measure *B* using a (calibrated) Hall probe **and** adjust / rotate probe until maximum value<br>**or**<br>measure *B* using Hall probe first in one direction, then in the opposite direction and average | |
| | D7 use a (top-pan) balance to measure *m* | |
| | D8 use of strong magnets to increase *v* | |
| | D9 repeat measurements of *v* for each value of *s* **and** average *v* | |
| | D10 relationship valid if a straight line is produced (passing through $\left(-\frac{2DQ}{m}\right)$)<br>Do not accept line passing through the origin. | |

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